

VYNGO TAKE-HOME

# Vivi

## Voice search for used commercial vehicles

A working agent that keeps every vehicle fact tied to the catalog.

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# The demo starts easy, then tests memory

1

## Find

Chhota truck chahiye, 5 lakh ke andar, city delivery ke liye.

2

## Correct

Nahi, diesel nahi, CNG.

3

## Ask

Second one ka payload aur GVW kya hai?

4

## Push it

Actually, show only CNG tankers under ₹1 lakh in Kolkata.

### Vivi

Commercial vehicle assistant

📖 Catalog connected

🗣️ Voice & Text Available

+ Start new chat

#### Current search

Budget, location and vehicle needs will stay visible here.

#### Conversation totals

Time, tokens and estimated cost will accumulate here.

#### Latest turn timing

Stage timings appear after the first turn.

#### Latest turn usage

Token usage appears after the first turn.

### Find the right commercial vehicle

Deploy ⋮

Natural English or Hinglish, grounded in the vehicle catalog.



Hi, I'm Vivi. Tell me what you need to carry, where you operate, and your budget. I'll help you find suitable commercial vehicles.

Not sure where to begin? Try one of these:

I need a small truck for city deliveries under ₹8 lakh

Which diesel trucks can carry at least 2 tonnes?

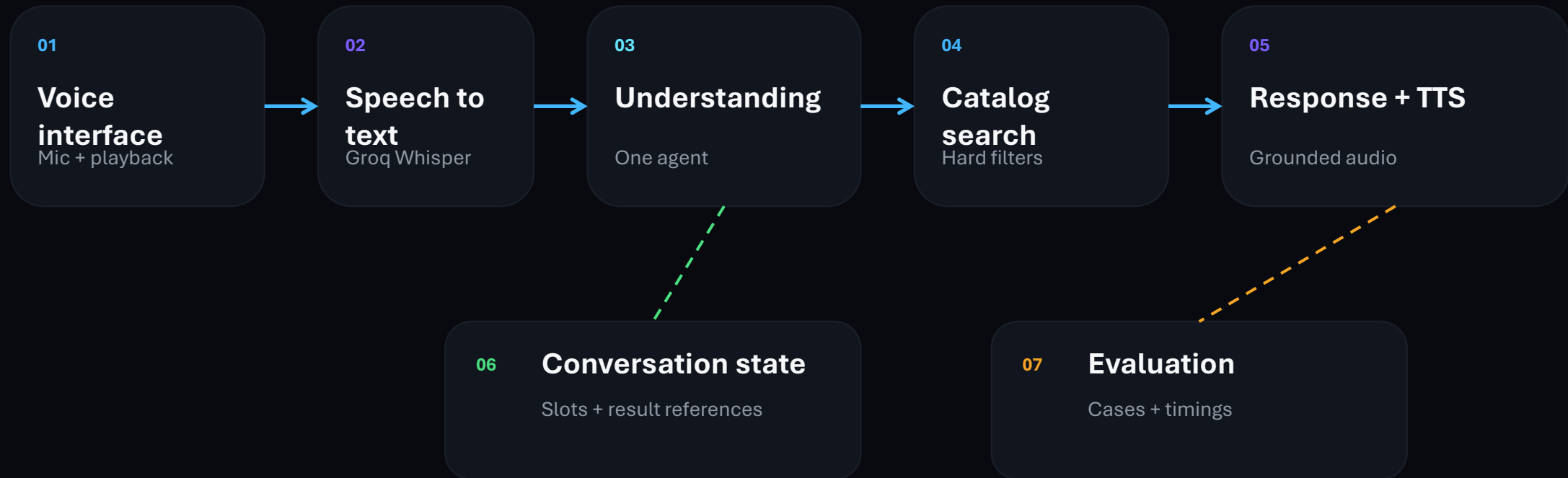
What commercial vehicles are available in Mumbai?

Describe a vehicle need or ask a follow-up



# Seven seams, one process

The live path stays short. State and evaluation observe it without becoming extra agents.



Every seam has a current implementation and a clean production replacement.

# ₹5 lakh means ₹5 lakh

Ranking can reorder eligible rows. It cannot rescue a row that breaks a constraint.



No match? Vivi tests one constraint at a time and names the first relaxation that produces a valid row.

## Ranking weights



Unused signals drop to zero. The remaining weights are normalized.

# Where the model stops

## MODEL

- Choose one typed tool
- Extract slot changes
- Phrase a detail answer when useful

## APPLICATION

- Build SQL and enforce filters
- Rank eligible vehicles
- Validate IDs and numeric facts
- Use deterministic fallback

**If a generated answer changes a listing ID or number, Vivi discards it and shows the deterministic answer.**

# Each turn changes only what the buyer changed

| TURN | WHAT THE BUYER SAYS                        | WHAT VIVI DOES                                  |
|------|--------------------------------------------|-------------------------------------------------|
| 01   | Small truck under 5 lakh for city delivery | <b>Sets category, budget and purpose</b>        |
| 02   | Nahi, diesel nahi, CNG                     | <b>Changes fuel only</b>                        |
| 03   | Second one ka payload aur GVW kya hai?     | <b>Reads two fields from saved result 2</b>     |
| 04   | CNG tanker under ₹1 lakh in Kolkata        | <b>Reports zero results and keeps prior IDs</b> |

A failed search changes the answer. The last valid result set remains available for later references.

# 51 of 52 live cases passed

## 98.1%

overall across three suites

**The failed case stayed in the score.**

Core conversation

28/28



Vehicle variants

18/18



Recorded voice

5/6



**Known miss**

STT and extraction produced "Bangalore" while the catalog expects "Bengaluru". The exact filter returned no result.

# The voice loop averages 5.32 seconds

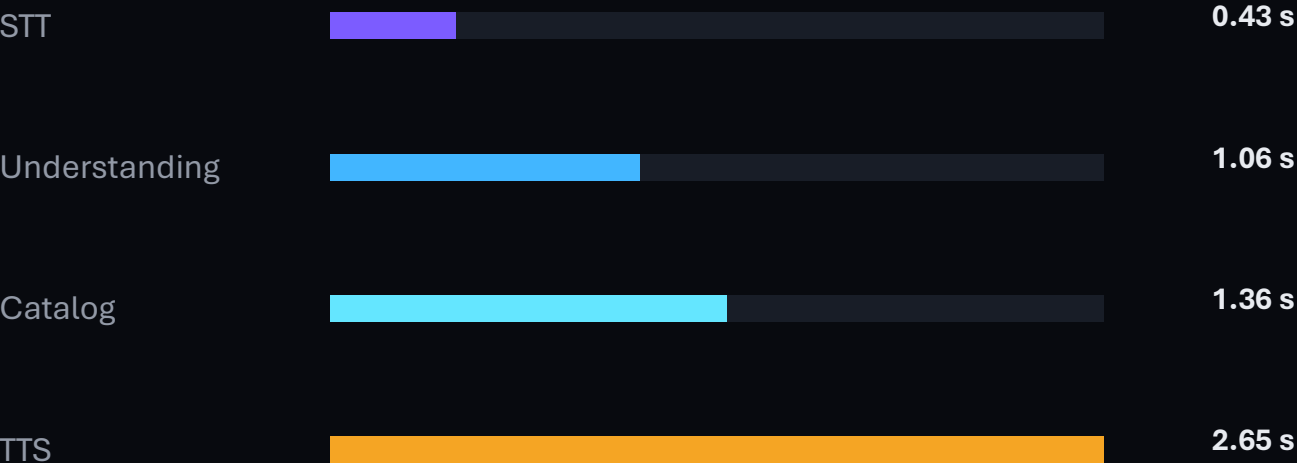
5.32 s

recording received to complete audio ready

₹0.4313

mean estimated list cost

## Mean stage timings



**TTS is the largest mean stage.**

The first catalog request also carried a 5.00 s cold connection.

Boundary: completed upload at the server to complete WAV ready. Browser recording and playback sit outside it.



# Four choices I would defend

| AREA     | CHOSEN                      | REJECTED                  | WHY                                            |
|----------|-----------------------------|---------------------------|------------------------------------------------|
| Search   | Typed tools + app SQL       | Model-written SQL         | Hard filters stay visible and testable         |
| Agent    | One agent, three tools      | Planner and critic chain  | Fewer calls and no routing disagreement        |
| Voice    | STT, text agent, TTS        | Speech-to-speech          | Transcript, slots and timings stay inspectable |
| Database | Scoped read-only connection | Process-global connection | Safe ownership across Streamlit sessions       |

# What breaks first at 100,000 conversations a month?

## Provider capacity.

Free-tier quotas and long-tail model latency fail before the 1,000-row catalog.

1 **Buy defined capacity and add circuit breakers**

2 **Stream STT and TTS**

3 **Move session state to Redis**

4 **Pool catalog access and cache stable options**

5 **Pin models in CI and run provider canaries**

# Seven component decisions

| COMPONENT       | WHY THIS CHOICE         | COMPLEXITY HIDDEN                 | PRODUCTION SEAM               |
|-----------------|-------------------------|-----------------------------------|-------------------------------|
| Voice interface | Fast demo surface       | Mic capture, upload, playback     | Product client with streaming |
| Speech to text  | Fits file input         | Codecs, retry, transcription      | Streaming STT + confidence    |
| Understanding   | Typed tool contracts    | Intent, slots, repair, routing    | Capacity-backed model         |
| Catalog search  | Strict query ownership  | Filters, ranking, read access     | Indexed service + pool        |
| Response + TTS  | Facts remain controlled | Validation and audio batching     | Streaming response gateway    |
| State           | Simple typed memory     | Corrections and result references | Redis or concurrent store     |
| Evaluation      | Behavior over prose     | Expected filters and telemetry    | Pinned CI + production traces |

These are seams, not separate services. The current build remains one process.

# What the evaluation actually checks

28

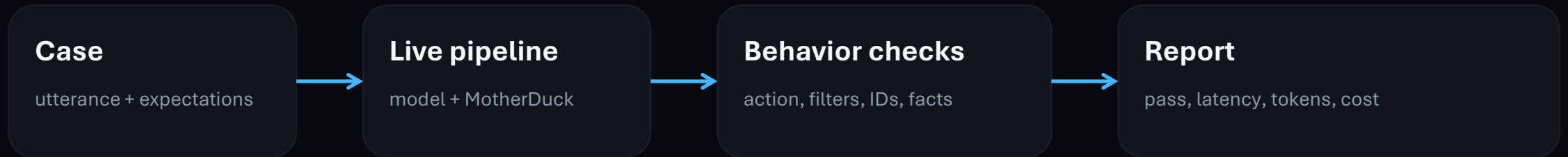
core

18

variants

6

voice

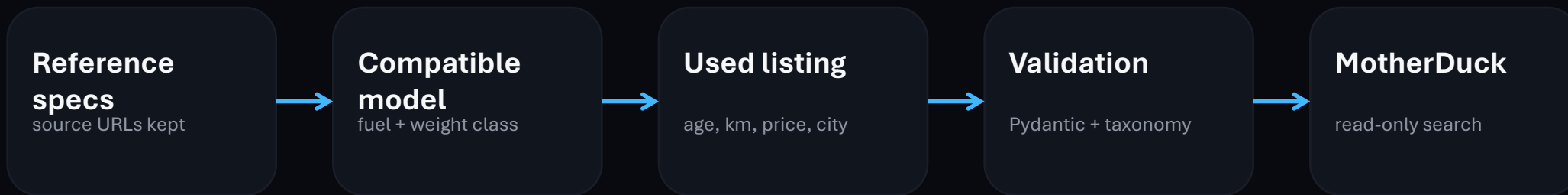


**A pass does not depend on one exact sentence. It depends on the executed behavior and catalog facts.**

# How the synthetic catalog stays believable

# 1,000

listings generated with seed 42



**Vehicle specifications stay attached to compatible models.**

Age, mileage, price, city, condition and verification remain synthetic demonstration data.

**Estimated payloads stay labeled.**

# The questions I would ask this system

|    |                                             |                                                                                                  |
|----|---------------------------------------------|--------------------------------------------------------------------------------------------------|
| 01 | <b>STT hears "S" instead of "Ace"</b>       | Exact filtering returns no match. Add catalog-aware normalization with a confirmation threshold. |
| 02 | <b>Every provider returns 429</b>           | The UI shows a recoverable error. Buy capacity and add circuit breakers first.                   |
| 03 | <b>Two hundred sessions arrive together</b> | Move SQLite state to Redis and replace repeated database setup with a bounded pool.              |
| 04 | <b>Where can 200 ms disappear?</b>          | Warm catalog connections. Requested-field follow-ups can also skip natural rephrasing.           |
| 05 | <b>Why did these three rank first?</b>      | Open the numeric score breakdown. Purpose fit carries 30 percent when the query includes it.     |

The code is ready for each answer.